CSA1735-ARTIFICIAL INTELLIGENCE

CO1 – AT1

ANALYTICAL PROBLEM SOLVING

Assessment: Questions 1, 9, 10, 12 and 13

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1. BRIDGE AND TORCH PROBLEM (STATE-SPACE FORMULATION AND UCS)

Problem: Four people need to cross a bridge at night. They have one torch and no more than two people can cross at once. The crossing times are 1, 2, 5 and 10 minutes. When two people cross together, their crossing takes the slower person's time. Find the minimum total time.

State-space formulation

- State: (L, T), where L is the set of people on the left side and T is the torch position (Left or Right). Everyone not in L is on the right side.
- Initial state: All four people and the torch are on the left.
- Goal state: All four people are on the right, with the torch on the right.
- Actions: One or two people on the same side as the torch cross to the opposite side carrying the torch.
- Transition model: The selected person or pair and the torch change sides.
- Path cost: A single person takes their own time. A pair takes the larger of their two times. Total path cost is the sum of crossing times.
- Search: Uniform Cost Search expands the state with the lowest accumulated cost g(n). With non-negative costs, it returns an optimal path.

Optimal crossing sequence

Step	Movement	Cost	Total
1	1 and 2 cross to the right	2 min	2 min
2	1 returns to the left	1 min	3 min
3	5 and 10 cross to the right	10 min	13 min
4	2 returns to the left	2 min	15 min
5	1 and 2 cross to the right	2 min	17 min

Result: The minimum total crossing time is 17 minutes.

9. SMART TRAFFIC SIGNAL CONTROLLER (AGENT DESIGN)

Objective: Control traffic lights intelligently to reduce waiting time while maintaining road-user safety.

Agent design

- Agent type: Model-based and utility-based agent.
- Percepts: Vehicle counts, queue lengths, cameras, pedestrian requests, current signal phase, and emergency-vehicle detection.
- Environment: Road intersection, vehicles, pedestrians, and surrounding traffic conditions.
- Actions: Set North–South green, set East–West green, activate pedestrian crossing, extend a green phase, or safely prioritize an emergency vehicle.
- Performance measure: Lower average waiting time and queue length, higher traffic flow, safe pedestrian crossing, and no conflicting green signals.
- Constraints: Minimum and maximum green times, amber light, all-red clearance time, pedestrian crossing time, and safety rules.

Working procedure

1. Read traffic and pedestrian sensors.
2. Estimate queue lengths and identify emergency requests.
3. Select a signal phase that is safe and reduces delay.
4. Apply amber and clearance intervals before switching phases.
5. Activate the selected phase and monitor the results.
6. Repeat continuously as traffic conditions change.

Pseudocode

```
WHILE the controller is operating:
    read sensor data
    estimate queue lengths
    IF an emergency vehicle is detected:
        select a safe emergency-priority phase
    ELSE:
        select the safest phase with highest traffic demand
    enforce timing and safety constraints
    apply amber and clearance intervals
    activate the selected signal phase
```

Expected result: The controller adapts signal timing to traffic demand while keeping safety constraints mandatory.

10. PROBLEM FORMULATION: TOWER OF HANOI

Problem: Move a stack of n disks from peg A to peg C using peg B as an auxiliary peg. Only one disk may be moved at a time, and a larger disk cannot be placed on a smaller disk.

Problem formulation

- State: The position of every disk on the three pegs.
- Initial state: All disks are on peg A, largest at the bottom and smallest at the top.
- Goal state: All disks are on peg C in the same order.
- Actions: Move the top disk from one peg to another.
- Legal rule: A larger disk must never be placed on a smaller disk.
- Transition model: A legal move changes the location of one top disk.
- Path cost: Each move costs 1; the goal is to minimize the number of moves.

Recursive algorithm

```
HANOI(n, source, auxiliary, destination):
    IF n == 1:
        move disk from source to destination
        RETURN
    HANOI(n-1, source, destination, auxiliary)
    move disk from source to destination
    HANOI(n-1, auxiliary, source, destination)
```

Minimum number of moves: $T(n) = 2^n - 1$.

Example: For 3 disks, the minimum number of moves is $2^3 - 1 = 7$.

12. MARS ROVER AGENT (PEAS AND ARCHITECTURE)

Objective: Explore Mars, collect scientific information, avoid hazards, conserve energy, and transmit useful data to Earth.

PEAS description

PEAS element	Description
Performance measure	Scientific discoveries, safe navigation, mission completion, distance explored, energy efficiency, and successful data transmission.
Environment	Martian rocks, slopes, craters, dust, temperature changes, lighting, limited energy, and delayed communication with Earth.
Actuators / actions	Drive wheels, steering, robotic arm, drill or sample collector, camera positioning, instruments, and communication transmitter.
Sensors / percepts	Navigation cameras, hazard cameras, wheel odometry, inertial sensors, temperature sensors, sun sensors, and scientific instruments.

Agent architecture

- Perception: Collect camera, navigation, and instrument data.
- State estimation: Estimate location, orientation, hazards, energy, and mission status.
- Planning: Choose a safe route or scientific task based on goals and available resources.
- Control: Send commands to wheels, robotic arm, and instruments.
- Monitoring: Detect hazards or faults and stop or re-plan when needed.
- Communication: Store data and transmit it to Earth when a link is available.

Agent type: A model-based, goal-based agent with utility-based planning. It needs an internal model because the environment is only partly observable and communication is delayed.

Example: When a camera detects a steep slope, the rover updates its map, evaluates a safer route, checks energy requirements, and proceeds only if the route is within operating limits.

13. 8-QUEENS PROBLEM (FORMULATION AND SEARCH)

Problem: Place eight queens on an 8 × 8 chessboard so that no two queens attack each other. Queens attack along rows, columns, and diagonals.

Search formulation

- State representation: Q[c] = r means the queen in column c is placed in row r.
- Initial state: An empty chessboard.
- Successor function: Place one queen in the next column in a row that is not attacked.
- Goal test: Eight queens are placed with no shared row, column, or diagonal.
- Path cost: Each placement can be treated as unit cost; the aim is to find a valid arrangement.
- Search strategy: Backtracking depth-first search. If a placement causes conflict, remove it and try another row.

Conflict condition

Two queens at (r1, c1) and (r2, c2) conflict if r1 = r2 or |r1 - r2| = |c1 - c2|. Placing one queen per column automatically prevents column conflicts.

Backtracking pseudocode

```
SOLVE(column):
  IF column == 8: RETURN SUCCESS
  FOR each row from 0 to 7:
    IF row and diagonals are safe:
      place queen at (row, column)
      IF SOLVE(column + 1) succeeds: RETURN SUCCESS
      remove queen
  RETURN FAILURE
```

One valid solution, giving the row for columns 1 through 8, is [1, 5, 8, 6, 3, 7, 2, 4].

Conclusion: Backtracking solves the 8-Queens problem by checking constraints after each placement and abandoning partial arrangements as soon as a conflict is found.

Overall Conclusion

These questions apply major AI concepts: optimal state-space search, intelligent agent design, recursive problem solving, PEAS-based agent analysis, and constraint satisfaction using backtracking.